Visualizing Housing Market Trends: An Analysis of Sale Prices and Features using Tableau

TEAM MEMBERS:-

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ABOUT PROJECT:-

1. **Problem Statement:-** The primary challenge addressed by this project is the difficulty in interpreting vast, raw housing market datasets. Without proper visualization, it is nearly impossible for stakeholders to quickly identify trends in pricing, sales volume over time, or the impact of property age on marketability.

2. **Data Collection:-** We identified and collected raw housing market datasets, which were then successfully loaded into our processing environment for further analysis.

3. Data Cleaning and Preparation:- Raw data often contains inconsistencies. Our preparation phase involved:

- Formatting the dataset for optimal structure in Tableau.
- Handling missing data points and addressing outliers.

- Transforming variables to ensure the visualizations accurately represent market reality.

4. Visualizations and Dashboards:- Using Tableau, we developed:

- **Interactive Dashboards:** Featuring responsive graphs that react to user-applied filters.
- **Comprehensive Graphs:** Including Sales by Year, House Age Distribution (Pie and Bar charts), and Renovation impacts.
- **Storyboard Design:** A step-by-step data narrative that leads the user from initial observations to final conclusions.

5. Performance Testing:- To ensure the tool is production-ready, we tested:

- The amount of data successfully loaded.
- The utilization and responsiveness of data filters.
- Rendering speeds for multiple simultaneous visualizations.

6. Results and Insights:- The project resulted in a functional web-integrated application using Flask that hosts our Tableau findings. Key results include the ability to visualize house age distributions and track total sales trends across different years, providing a clear picture of market health and property lifecycles.

PROJECT EXPLANATION LINK:-
https://drive.google.com/file/d/1id-v9ZWlOlMCv0r4j10s2jBQLI8642Wm/view?usp=sharing

PORJECT HOSTED LINK:-
https://akshatverma79.github.io/Visualizing-Housing-Market-Trends/Web%20Integration/index.html